

Yourui Shao

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Education

California Institute of Technology

GPA: 4.13

B.S. in Computer Science (Learning & Vision), minor in Neurobiology

June 2028

- Courses: Algorithmic Decision-Making, Large Language Models, Machine Learning & Data Mining, Deep Learning, Learning Systems, Computing Systems, Software Design in C, Linear Algebra, Multivariable Calculus, Differential Equations, Probability & Statistics, Combinatorial Analysis

Experience

Undergraduate Researcher, Caltech Vision Lab

Sep 2026 – Present

Undergraduate Researcher, Caltech Rigorous Systems Research Group

Mar 2026 – Present

- Optimizing multi-agent systems inference-time scaling of general problem-solving domains with information theory.
- Developed custom pipeline for domain-specific skills template generation & embeddings-based retrieval to create subdomain experts for math problem solving. Analyzing leverage and efficacy of template expertise.
- Designing & running experiments. Developed comprehensive evaluation modules for inference, template, & retrieval with ablationary baseline testing, template & expert contribution metrics, & LLM answer judging.

Summer Undergraduate Research Fellow, Stanford Neural Prosthetics Translational Lab

Jun 2026 – Aug 2026

- Constructed hypotheses & experiments to characterize and causally explain emergent geometric structure of hierarchical syntactic, lexical, and phonemic speech representations in the brain.
- Trained, perturbed, & interpreted task-driven RNN models as analogs to neural population activity.
- Tested effects of task settings and model parameters. Demonstrated (1) weight decay regularization effects on queue dynamics observed in neural data; (2) emergence of grammar encoding as overlapping compressed word identities under regularization; and (3) shared neural subspace facilitating whole-word lexical → phoneme conversion.
- Developed toolkit to characterize hidden state geometries with linear encoders/decoders, null-potent space decomposition, subspace overlap and trajectories, effective dimensionality, rotation and linearity of solutions, etc.

Projects

Human Latent Space | Representation learning, LLMs

Jun 2026 – Sep 2026

- Completed as researcher at small pre-seed startup for modeling human interaction and social dynamics.
- Showed that self-supervised learning per-individual context embeddings to optimize personalized conversation simulation with Qwen simultaneously yields representations encoding speaker personality traits (CANDOR dataset).
- Used single gradient step learning to demonstrate compatibility of objectives. Created custom joint training optimization schedules, history ensembling and Transformer-free baselines, & controlled feature encoding/decoding tests. Troubleshoot mode collapse with alternative loss objectives (cosine similarity, KL divergence, unlikelihood).

Relay (HackPrinceton 2025) | Python, LangChain, FastAPI, TypeScript, React

Nov 2025

- In 20 hours, built functioning full-stack, multi-agent AI web application to look up and cite, write articles, and notify users on state legislative activity with self-updating RAG system live scraping congressional sites.
- Developed conversational agent with custom source retrieval tooling and adaptive to user interest and reading level.

Reinforcement Learning for Fish | Python, PyTorch

Jan 2025 – May 2025

- Taught RL to play *Fish*, a multiplayer, team-based card game. Built and optimized two-part convolutional LSTM hand prediction + Q-learning network in PyTorch inspired by human gameplay tactics. Handwrote all the code.
- Programmed custom state-action converter, data parser, self-play environment, data pipeline, & rewards.

Publications

Predicting College Admission Results with Machine Learning on Unstructured Online Data, 2024 IEEE FMLDS

John Tian, Yourui Shao

An Accurate Classification and Recommendation Method of Competitive Math Problems, 2024 IEEE BigDataService

Yourui Shao

Skills

- **Languages & Tools:** Python, LangChain, FastAPI, Flask, PyTorch, TensorFlow, Numpy, Pandas, Java, C, JavaScript, TypeScript, React, HTML/CSS, SQL (Postgres, MySQL, SQLite), git, Docker
- **Concepts:** Post-training, NLP, Transformers, RL, deep learning, multi-agent systems, REST APIs, full stack, databases